MIL-E-85246A(AS)
13 November 1984

Supersession Data
(See Section 6)

# MILITARY SPECIFICATION

# ELECTRONIC CIRCUIT MODULES

# GENERAL SPECIFICATION FOR

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification establishes, in conjunction with applicable assembly drawings for the specific type and characteristics of the item needed, the requirements for manufacture and acceptance of printed wiring assemblies, referred to herein as the "modules."

1.2 Classification. Modules described by this specification are fabricated for procurement at the module level (spares) and for procurement at a higher level of assembly. Modules described by this specification are classified as follows, depending on the physical construction of the module:

a. Type A: Conformally coated module.

b. Type B: Encapsulated conformally coated module.

This type distinction is solely for the purpose of internal construction inspection of the Type B module before encapsulation. All other requirements of this specification are the same for Type A and Type B modules.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1462) appearing at the end of this document or by letter.

FSC 1420

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## 2. APPLICABLE DOCUMENTS

### 2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-P-116 | Preservation, Methods of. |
| MIL-T-18303 | Test Procedures; Preproduction, Acceptance, and Life for Aircraft Electronic Equipment, Format for. |
| MIL-P-28809 | Printed Wiring Assemblies. |
| MIL-I-46058 | Insulating Compound, Electrical. |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-109 | Quality Assurance Terms and Definitions. |
| MIL-STD-129 | Marking for Shipment and Storage. |
| MIL-STD-130 | Identification Marking of U.S. Military Property. |
| MIL-STD-202 | Test Methods for Electronic and Electrical Component Parts. |
| MIL-STD-454 | Standard General Requirements for Electronic Equipment. |



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| MIL-STD-794 | Parts and Equipment, Procedures for Packaging and Packing of. |
| MIL-STD-831 | Test Reports, Preparation of. |
| MIL-STD-1695 | Environments, Working, Minimum Standards for. |
| MIL-STD-45662 | Calibration Systems Requirements. |

2.1.2 Other Government documents and publications. The following other Government documents and publications, of the issue specified in the solicitation, form a part of this specification to the extent specified herein.

OTHER GOVERNMENT DOCUMENTS

Code of Federal Regulations (CFR)

| | |
|---|---|
| 49 CFR 171-178 | Transportation. |

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

PUBLICATIONS

Naval Air Systems Command
(Code Ident 30003)

| | |
|---|---|
| AD-1322 | Reliability Program Provisions for Production of Sidewinder AIM-9. |
| WS-6536 | Procedures and Requirements for Preparation and Soldering of Electrical Connections. |

(Copies of specifications, standards, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.



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(fire retardant curtains) at the following locations:

## 3. REQUIREMENTS

3.1 Item description. Modules covered by this specification consist of two packaging configurations of electronic components and single function input-output terminal leads assembled to printed wiring boards using soldered connections. The Type A configuration is that of rigid multilayer printed wiring boards and electronic components conformally coated with an insulating compound to provide physical and environmental protection for the module assembly. The Type B configuration is that of conformally coated, rigid, double sided printed wiring board and electronic components encapsulated in a vulcanizing silicone rubber compound to provide physical and environmental protection for the module assembly.

3.2 Characteristics.

3.2.1 Performance.

3.2.1.1 Electrical. The electrical characteristics of the module shall be as specified on the applicable drawing.

3.2.1.1.1 Post-environmental electrical. Subsequent to burn-in or environmental exposure, the module's ambient temperature electrical characteristics shall meet the pre-environmental requirements of the applicable drawing.

3.2.1.1.2 High temperature electrical. The module's high temperature electrical characteristics shall be in accordance with the post-environmental requirements of the applicable drawing, unless specific high temperature requirements are specified by the drawing.

3.2.1.1.3 Life test electrical. During and subsequent to life testing, the module's ambient temperature electrical characteristics shall meet the post-environmental requirements of the applicable drawing.

3.2.1.2 Environmental. The module shall withstand the following environments without impairment to the conformal coating or encapsulant in accordance with 3.3.3. The electrical characteristics of the module subsequent to environmental exposure shall be in accordance with 3.2.1.1.1.

3.2.1.2.1 High temperature operation. The module shall meet the high temperature electrical requirements of 3.2.1.1.2 while operating at 90 ±3 degrees Celsius (°C).



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3.2.1.2.2 Thermal shock. The module shall withstand thermal shock as follows:

a. During first article testing, each module shall be subjected to the thermal shock in accordance with MIL-STD-202, Method 107, Test Condition A-2.

b. During quality conformance inspection, each module shall be subjected to the thermal shock in accordance with MIL-STD-202, Method 107, Test Condition A.

3.2.1.2.3 Mechanical shock. The module shall withstand mechanical shock in accordance with MIL-STD-202, Method 213, Test Condition A.

3.2.1.2.4 Random vibration. The module shall withstand random vibration in accordance with Figure 1 for a minimum duration of 2 hours in each axis. The test shall be conducted in accordance with MIL-STD-202, Method 214.

3.2.1.2.5 Humidity. The module shall withstand the humidity conditions in accordance with MIL-STD-202, Method 103, Test Condition B.

3.2.1.3 Physical characteristics.

3.2.1.3.1 Terminal strength. Each module terminal shall withstand a 5.0-pound axial pull in accordance with MIL-STD-202, Method 211, Test Condition A. Subsequently, the module shall exhibit no visual evidence of terminal breakage, terminal loosening, or cracking of the module surface. Deformation of preformed leads of the Type A module resulting from the terminal strength test shall not constitute a failure.

3.2.1.3.2 Solderability. Each module lead shall meet the solderability requirements of MIL-STD-202, Method 208.

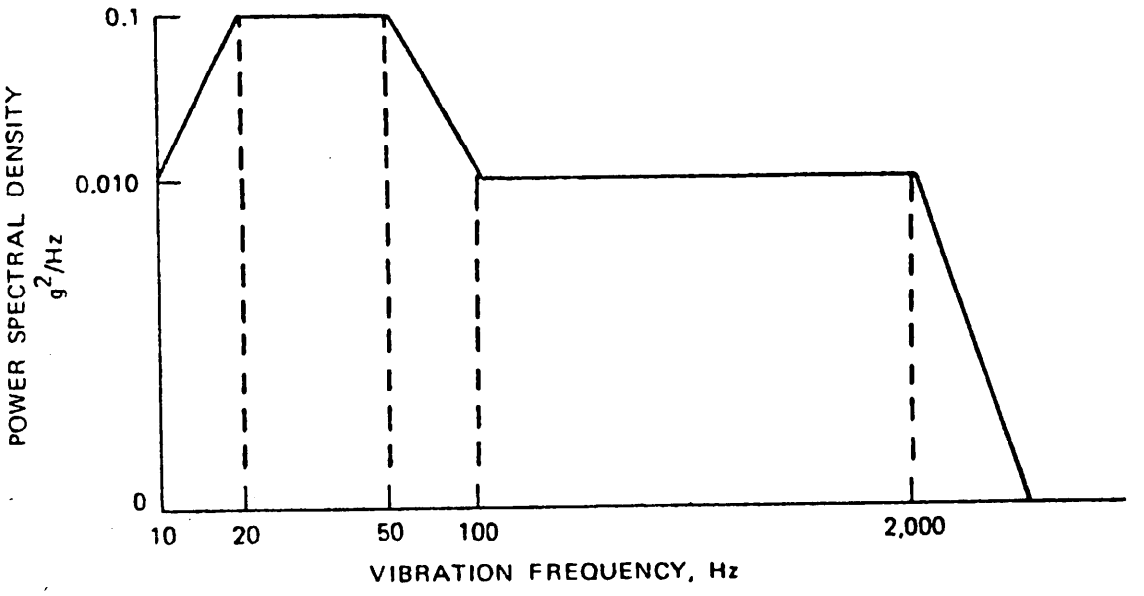
3.2.1.4 Life expectancy. The module shall exhibit a minimum life of 500 hours of operating time under the test conditions in accordance with Figure 2. The module electrical characteristics throughout its simulated life shall be in accordance with 3.2.1.1.3.

3.3 Design and construction. The design, construction, materials, and finish of the module shall be such that the module will meet all requirements specified herein. Acceptance or approval of any such constituent shall not be construed as a guarantee of acceptance of the finished product.

3.3.1 Production drawings. The module shall be manufactured and assembled in accordance with the applicable drawings and as specified herein.



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NOTES:

1/ SPECTRAL DENSITY VALUES SHALL BE ±1.5 dB OF THE SPECIFIED VALUES BETWEEN 10 TO 1,000 Hz AND ±3 dB BETWEEN 1,000 TO 2,000 Hz

2/ SPECTRAL DENSITY CURVE END POINT ROLL-OFFS SHALL BE NO LESS THAN 10 dB/OCTAVE

FIGURE 1. Random vibration test curve envelope.



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3.3.1.1 Selection of parts and materials. Parts and materials not specified in this specification shall be selected in accordance with AD-1322.

3.3.1.2 Conformal coating. All modules shall be conformally coated with an insulating compound in accordance with MIL-I-46058. The coating shall be applied to both sides of the cleaned printed wiring assembly, including the components and their leads and all exposed conductors except heat sinks.

3.3.1.3 Potting compound. Type B modules only shall be encapsulated in a vulcanizing silicone rubber that conforms to the applicable drawing.

3.3.1.4 Cleaning solvent. The solvent used for cleaning or preparation of modules for precoating, encapsulation, or conformal coating shall be in accordance with WS-6536.

3.3.1.5 Cleanliness. The module shall pass the cleanliness requirements of MIL-P-28809 prior to encapsulation or conformal coating.

3.3.1.6 Mold release agent. Mold release agents shall be compatible with the encapsulant, and shall not result in peeling, flaking, or crazing of the module surface. The residual mold release agent shall be removed from the surfaces of the module.

3.3.1.7 Other materials. Any other materials used in the fabrication of the module shall meet the applicable requirements of MIL-STD-454, Requirement 3, Flammable Materials; Requirement 4, Fungus-Inert Materials; Requirement 15, Ferrous Alloys, Corrosion Resistance; and Requirement 16, Dissimilar Metals.

3.3.2 Standards of manufacture.

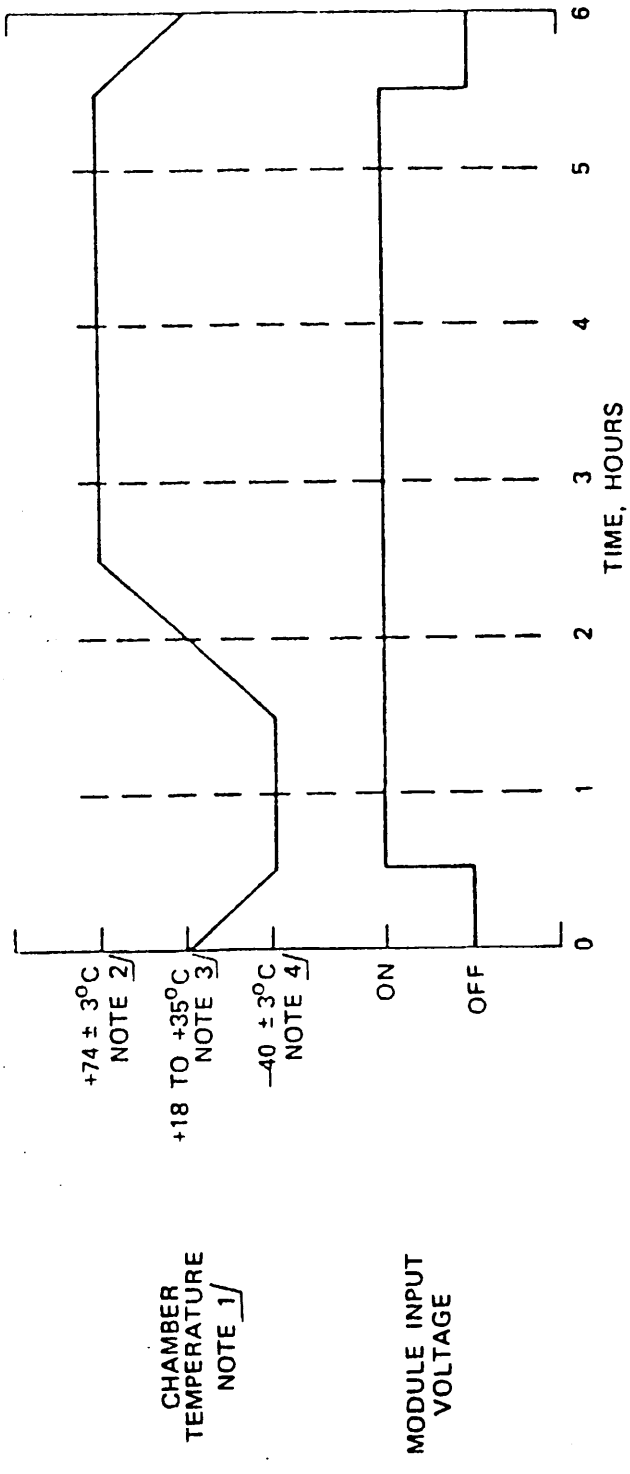
3.3.2.1 Special working conditions. The facilities used in the manufacture, testing, and inspection of the modules shall be in accordance with the provisions of MIL-STD-1695.

3.3.2.2 Soldering. Procedures and requirements for preparation, assembly, and soldering of electrical connections shall be in accordance with WS-6536.

3.3.3 Appearance. When the module is tested as specified herein, the conformal coating and encapsulant (as appropriate) shall remain free from discoloration, blistering, wrinkling, cracking, and peeling. Bubbles and pinholes in conformal coating on top of printed wiring board conductors shall not be permitted. Bubbles in open areas of the printed wiring board greater than 0.025 inches shall not be acceptable. Bubbles adjacent to electrical components, which are greater than 0.010 inches in diameter, shall not be acceptable. Bubble and micro bubble clusters



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NOTES:

1/ CHAMBER TEMPERATURE RATE OF CHANGE SHALL BE <3°C/MINUTE

2/ MODULE SOAK TIME AT +74°C SHALL BE 3 HOURS, +15 MINUTES, −0 MINUTES

3/ AMBIENT TEMPERATURE TEST MEASUREMENT RANGE

4/ MODULE SOAK TIME AT −40°C SHALL BE 1 HOUR, +15 MINUTES, −0 MINUTES

FIGURE 2. Module life test temperature/power test cycle.



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(porosity) in open areas near printed wiring board edges shall be not greater than 0.26 inches in length, or when adjacent to a components's side, shall be not greater than 25 percent of that side in length. A cluster of bubbles between two circuit paths shall not form a bridge where the clearance between any two bubbles or a bubble and a circuit is less than the minimum spacing requirements of the drawing. Voids shall not be acceptable. The conformal coating shall not mask or obliterate the identification markings and color codes of electronic components.

3.4 Special processing.

3.4.1 Burn-in. Each module shall be subjected to six cycles of burn-in with thermal cycling in accordance with Figure 3.

3.5 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspections (see 4.7 and 6.2.3).

3.5.1 First article verification. Unless otherwise specified in the contract or purchase order (see 6.2.3), the contractor shall perform first article verification (resubmission for first article testing) on any module identified by a unique part number during the production phase under any of the following conditions:

a. When the contractor has modified any module through a change of raw materials, processes, production procedures, or methods. It shall be the responsibility of the contractor to notify the procuring activity prior to the incorporation of any such changes and to provide quantitative evidence of the affect of such changes on the performance or characteristics of the module. The requirements for such tests shall be based on an evaluation of the evidence supplied.

b. Where there is evidence that the quality level of any module has not been maintained. This evidence may be in the form of accumulated failure reports of a module, system failures attributable to a module, or failure of a module to pass any of the tests that may be conducted by or for the procuring activity.

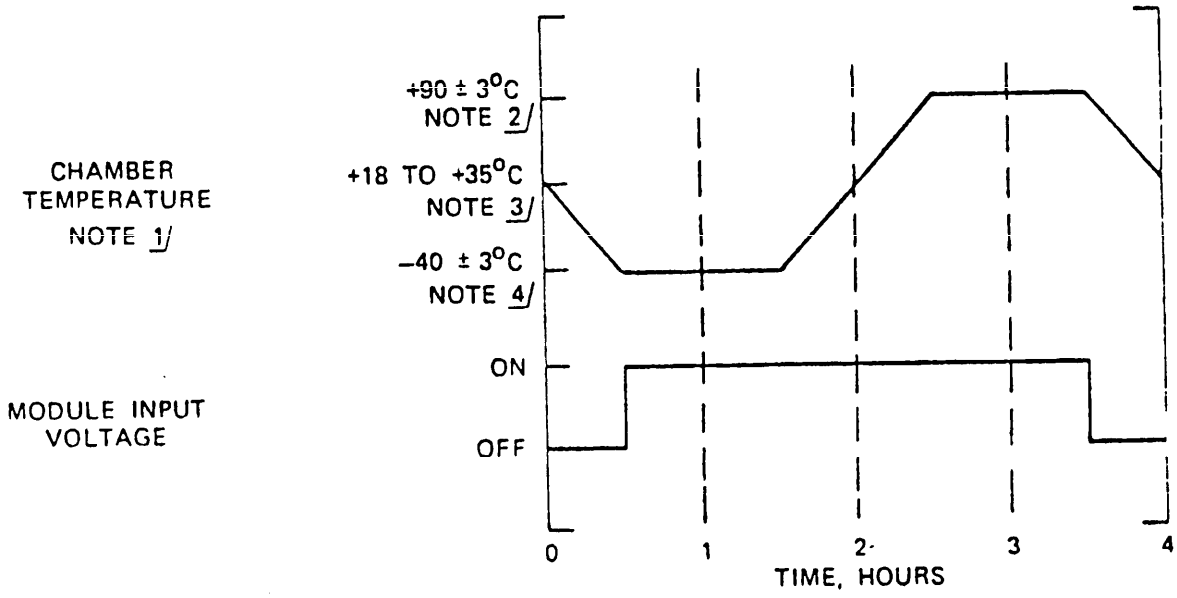
c. When any change in module design or documentation by the procuring activity affects previous first article baselines.

d. When any module is out of production for any period exceeding 18 months.

Exemption from first article verification for any module will be at the option of the procuring activity and will be determined on a case-by-case basis.



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NOTES:

1/ CHAMBER TEMPERATURE RATE OF CHANGE SHALL BE $< 3^{\circ}C$/MINUTE

2/ MODULE SOAK TIME AT $+90^{\circ}C$ SHALL BE 1 HOUR, +15 MINUTES, –0 MINUTES

3/ AMBIENT TEMPERATURE TEST MEASUREMENT RANGE

4/ MODULE SOAK TIME AT $-40^{\circ}C$ SHALL BE 1 HOUR, +15 MINUTES, –0 MINUTES

FIGURE 3. Module burn-in temperature/power test cycle.



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3.6 Documentation. When specified in the contract or purchase order (see 6.2.2), the following documentation shall be prepared:

a. Test procedures (see 4.3).

b. Certification (see 4.5.1).

c. Test station certification (see 4.5.4).

d. Test plans (see 4.7.1).

e. Test reports (see 4.7.5).

f. Failure summary report (see 4.11).

3.7 Workmanship. Fabrication and finishing shall be accomplished in a manner that will ensure freedom from warps, burrs, projecting ridges, cracks, chipped edges or surface blisters, uneven surfaces, dents, and scratches. Type A modules shall not have damaged or bent terminals.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspections specified herein are classified as follows:

a. First article inspections (see 4.7).

b. Quality conformance inspections (see 4.8).

4.3 Test procedures. When specified in the contract or purchase order (see 6.2.2), prior to performance of any tests specified herein, detailed test procedures for each module conforming to the applicable requirements of this specification and MIL-T-18303 shall be prepared by the contractor and shall be subject to approval by the procuring activity. All testing shall be performed in accordance with the accepted procedures, and any deviations or changes require prior approval of the procuring activity.



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4.4 Test conditions. Unless otherwise specified herein, all measurements shall be made under the following environmental conditions:

| | | |
|---|---|---|
| a. | Altitude: | Normal elevation of test facility. |
| b. | Ambient temperature, operating: | 18 to 35°C. |
| c. | Relative humidity: | Ambient (not exceeding 95 percent). |
| d. | Vibration: | Normal condition of test facility. |

4.5 Test equipment.

4.5.1 Calibration. All measuring and test equipment utilized by the contractor for performance of the inspections specified herein shall be calibrated and maintained in accordance with a system meeting the requirements of MIL-STD-45662 by using secondary laboratory standards for determination of accuracy. The secondary laboratory standards shall have been calibrated against primary standards with recalibration on a regularly scheduled basis. The referred source for the primary standards shall be certified by the National Bureau of Standards. The contractor shall certify that all test equipment has been so calibrated before testing begins. When specified in the contract or purchase order, certification to this effect shall accompany all reports of first article and quality conformance inspections (see 6.2.2).

4.5.2 Test equipment selection. All electrical and electronic measuring equipment used for module testing shall be as specified on the applicable drawing. Exceptions or substitutions to the test equipment list on the applicable drawing shall require approval by the cognizant design activity prior to use for any testing specified herein. Test equipment recommended as substitutions to those specified on the applicable drawing shall be of such a nature and design that any errors produced do not exceed 15 percent of the total tolerance allowed when both upper and lower limits are given, or 10 percent of the tolerance when only one limit is given.

4.5.3 Test chamber. In addition to the basic test chamber requirements given in MIL-STD-202, the contractor shall be responsible for selecting a chamber design and module loading configuration that subjects each module tested or inspected to the following conditions:

4.5.3.1 Temperature uniformity. The air temperature throughout the chamber working space shall be within ±3°C of the temperature at the chamber-control reference point. The test of 4.10.5 shall be used to verify compliance with this condition.

4.5.3.2 Module mounting. Each module shall be mounted in a manner which simulates service-use application.



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4.5.3.3 Radiation. Each module shall be free of direct thermal radiation when employing radiant type heaters.

4.5.3.4 Module operation. Chambers used for the life tests (see 4.10.4.1) and burn-in (see 4.10.4.2) shall have provisions for operating each module in accordance with 4.10.4.1.2.

4.5.4 Test station certification. When specified in the contract or purchase order, all test stations used for final acceptance of modules shall require certification by the procuring activity (see 6.2.1 and 6.2.2).

4.6 Special working conditions. To meet the requirements of 3.3.2.1, the contractor shall provide adequate facilities for the fabrication, assembly, and inspection of modules to be delivered in accordance with this specification. Unless otherwise specified in the contract or purchase order, the minimum standards for working environments of MIL-STD-1695 shall apply (see 6.2.1).

4.7 First article inspections.

4.7.1 Test plans. When specified in the contract or purchase order (see 6.2.2), a detailed test plan for first article testing shall be prepared by the contractor and shall be subject to approval by the procuring activity prior to the commencement of testing.

4.7.2 First article sample. A first article sample shall be required for each module identified by a unique part number. The sample shall be fabricated in accordance with this specification and the applicable drawings, and shall use the same methods, materials, processes, and procedures proposed for production (see 6.2.3). Any procurement prior to approval of the first article shall be at the risk of the contractor. The first article sample for each module identified by a unique part number shall consist of the following:

a. Type A: Twelve conformally coated modules that successfully complete the quality conformance inspection of 4.8.

b. Type B: Two unencapsulated modules representative of production hardware and twelve encapsulated modules that successfully complete the quality conformance inspection of 4.8.

4.7.3 First article tests.

4.7.3.1 Type A modules. The conformally coated modules of the Type A first article sample group shall be subjected to the tests of the appropriate group of Table I in the sequence shown. The modules shall



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TABLE I. First article inspections.

| Inspection | Requirement paragraph | Method paragraph |
|---|---|---|
| Group I (2 unencapsulated modules) (Type B modules only) | | |
| Visual (internal) | 3.3.1, 3.3.2.2, 3.3.3, and 3.7 | 4.9.1.2 |
| Group II (5 modules) (Type A and B modules) | | |
| High temperature, operating | 3.2.1.2.1 and 3.2.1.1.2 | 4.10.3.1 |
| Thermal shock | 3.2.1.2.2.a | 4.10.3.2.a |
| Electrical | 3.2.1.1.1 | 4.10.1 |
| Mechanical shock | 3.2.1.2.3 | 4.10.3.3 |
| Electrical | 3.2.1.1.1 | 4.10.1 |
| Random vibration | 3.2.1.2.4 | 4.10.3.4 |
| Electrical | 3.2.1.1.1 | 4.10.1 |
| Humidity | 3.2.1.2.5 | 4.10.3.5 |
| Electrical | 3.2.1.1.1 | 4.10.1 |
| Terminal strength | 3.2.1.3.1 | 4.10.2.1 |
| Visual (external) | 3.3.1, 3.3.2.2 1/, 3.3.3, and 3.7 | 4.9.1.1 |
| Group III (2 modules) (Type A and B modules) | | |
| Solderability | 3.2.1.3.2 | 4.10.2.2 |
| Visual (external) | 3.3.1, 3.3.2.2 1/, 3.3.3, and 3.7 | 4.9.1.1 |
| Group IV (5 modules) (Type A and B modules) | | |
| High temperature, operating | 3.2.1.2.1 and 3.2.1.1.2 | 4.10.3.1 |
| Life | 3.2.1.4 | 4.10.4.1 |
| Electrical | 3.2.1.1.3 | 4.10.1 |
| Visual (external) | 3.3.1, 3.3.2.2 1/, 3.3.3, and 3.7 | 4.9.1.1 |

1/ Not applicable to Type B modules.



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be randomly separated into three subgroups. The first subgroup of five shall be subjected to the tests as specified in Table I, Group II. The second subgroup of two shall be subjected to the tests as specified in Table I, Group III. The remaining subgroup of five shall be subjected to the tests as specified in Table I, Group IV.

4.7.3.2 Type B modules. Each unencapsulated module of the Type B first article sample shall be subjected to the examinations of Table I, Group I. The encapsulated modules of the Type B first article sample group shall be subjected to the tests of the appropriate group of Table I in the sequence shown. The modules shall be randomly separated into three subgroups. The first subgroup of five shall be subjected to the tests as specified in Table I, Group II. The second subgroup of two shall be subjected to the tests as specified in Table I, Group III. The remaining subgroup of five shall be subjected to the tests as specified in Table I, Group IV.

4.7.4 First article rejection. The failure of any first article sample module to meet any of the requirements specified herein shall reject the first article.

4.7.5 Test reports. When specified in the contract or purchase order (see 6.2.1), test reports in accordance with MIL-STD-831 (see 6.4) shall be prepared by the contractor for all first article inspections (see 6.2.2).

4.7.6 Disposition of first article sample. Unless otherwise specified in the contract or purchase order (see 6.2.1), one-half of first article sample modules shall be retained by the contractor at the conclusion of the first article tests of 4.7.3. The remaining first article modules shall be shipped to the cognizant design activity at the conclusion of the first article tests.

4.8 Quality conformance inspections.

4.8.1 Individual inspections. Each module shall be subjected to the inspections and tests of Table II in the sequence shown. Any module that fails to pass any of the inspections and tests shall be rejected. Modules which have been reworked or repaired shall repeat the inspections and tests of Table II. No module shall pass to the next assembly level or be offered to the Government for acceptance (spares) without meeting all the requirements of Table II. The final electrical test results of acceptable modules shall be made into a permanent record and made readily available to the procuring activity upon request.

4.8.1.1 Individual tests (alternate). Prior to conformal coating or encapsulation, each module shall be subjected to the inspections and tests of Table III in the sequence shown. Any module that fails to pass any of the inspections and tests shall be rejected. Modules that are in need of rework or repair after being subjected to the initial electrical tests or high temperature testing shall repeat the inspections and tests of Table III starting with the first electrical test. No module shall pass to the next assembly level or be offered to the Government for acceptance



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(spares) without meeting all the requirements of Table III. The final electrical test results of acceptable modules shall be permanently recorded and made available to the procuring activity upon request.

4.8.2 Failure analysis and reporting. Failure analysis and reporting (see 4.11) shall be immediately initiated in the event of a failure of a module while undergoing the final electrical inspection of 4.10.1. Procuring activity approval shall be required for any corrective design changes.

TABLE II. Quality conformance inspections.

| Inspection | Requirement paragraph | Method paragraph |
|---|---|---|
| Thermal shock | 3.2.1.2.2.b | 4.10.3.2.b |
| Burn-in with thermal cycling | 3.4.1 | 4.10.4.2 |
| High temperature, operating | 3.2.1.1.2 and 3.2.1.2.1 | 4.10.3.1 |
| Electrical 1/ | 3.2.1.1.1 | 4.10.1 |
| Visual (external) | 3.3.1, 3.3.2.2 (Type A only), 3.3.3, and 3.7 | 4.9.1.1, 4.9.1.3, and 4.12 |

1/ A permanent record shall be retained and provided upon request.

4.9 Examination methods.

4.9.1 Visual examinations.

4.9.1.1 Visual examination (external). The module shall be examined for conformance to drawing requirements (see 3.3.1), standards of manufacture (see 3.3.2.2 for Type A only), standards of workmanship (see 3.7), and appearance (see 3.3.3).

4.9.1.2 Visual examination (internal Type B module only). The module shall be examined for conformance to drawing requirements (see 3.3.1), standards of manufacture (see 3.3.2.2), standards of workmanship (see 3.7), and appearance (see 3.3.3).

4.9.1.3 Materials certification. The contractor shall have written certification available for procuring activity inspection, accompanied by objective quality evidence (in accordance with MIL-STD-109), that the parts, materials, and processes conform to the requirements specified herein and on the applicable drawings.



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TABLE III. Quality conformance inspections (alternate).

| Inspection | Requirement paragraph | Method paragraph |
|---|---|---|
| Thermal shock | 3.2.1.2.2.b | 4.10.3.2.b |
| Electrical | 3.2.1.1.1 | 4.10.1 |
| Burn-in with thermal cycling | 3.4.1 | 4.10.4.2 |
| High temperature, operating 1/ | 3.2.1.1.2 and 3.2.1.2.1 | 4.10.3.1 |
| Electrical 2/, 3/, 4/ | 3.2.1.1.1 | 4.10.1 |
| Visual (external) | 3.3.1, 3.3.2.2 (Type A only), 3.3.3, and 3.7 | 4.9.1.1, 4.9.1.3, and 4.12 |

1/ Soak temperature shall be sufficient to stabilize module surface within 5°C of ambient temperature. Heat sinking/cooling shall be in accordance with assembly drawing.

2/ A permanent record shall be retained and provided upon request.

3/ Conformal coating or encapsulation shall be performed at completion of electrical requirements.

4/ Adequate temperature control shall be provided for modules containing hybrids to assure that temperature rise during test does not exceed 30°C.

4.10 Test methods.

4.10.1 Electrical tests. The module's electrical characteristics shall be determined in accordance with 3.2.1.1. Alternate but equivalent test circuits or procedures may be utilized if approved by the module cognizant design activity (see 6.3) and certified in accordance with 4.5.4.

4.10.2 Physical tests.

4.10.2.1 Terminal strength. Each module terminal (lead wire) shall be subjected to and shall pass a 5-pound pull test in accordance with MIL-STD-202, Method 211, Test Condition A (see 3.2.1.3.1).



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each module identified by a unique part number. The sample shall

4.10.2.2 Solderability. Each lead of the module shall be subjected to and shall pass the solderability test for solid or stranded wire leads in accordance with MIL-STD-202, Method 208 (see 3.2.1.3.2).

4.10.3 Environmental tests. Environmental tests shall be performed as specified herein. Unless otherwise specified herein, the test specimens shall be allowed to stabilize at room ambient conditions prior to performance of post-environmental tests and measurements. Upon completion of the environmental tests specified herein, the module shall be visually inspected for evidence of physical damage and deterioration due to exposure to the environments specified.

4.10.3.1 High temperature, operating. The module shall be temperature conditioned at 90 ±3°C for a minimum of 1 hour and then electrically tested, at temperature, in accordance with 3.2.1.1.2. The module may not be removed from the test chamber to perform the electrical tests (see 3.2.1.2.1). Modules completing the final 90°C period of burn-in (see 3.4.1) may be transferred within 5 minutes to the test chamber and temperature stabilized at 90 ±3°C and then electrically tested at temperature in accordance with 3.2.1.1.2.

4.10.3.2 Thermal shock. The module shall be subjected to the thermal shock test in accordance with MIL-STD-202, Method 107, and as follows:

a. For first article testing of modules, perform the test in accordance with Test Condition A-2.

b. For quality conformance inspection of modules, perform the test in accordance with Test Condition A.

In all cases, the module soak time shall be 30 minutes for steps 1 and 3 of MIL-STD-202 (see 3.2.1.2.2).

4.10.3.3 Mechanical shock. The module shall be mounted on a suitable test fixture and subjected to mechanical shock in accordance with MIL-STD-202, Method 213, Test Condition A. The module shall be mounted by a method which represents the in-service mounting and the shocks shall be applied along module axes which correspond to the missile vertical (umbilical), lateral, and longitudinal axes (see 3.2.1.2.3).

4.10.3.4 Random vibration. The module shall be mounted on a suitable test fixture and subjected to a minimum of 2 hours of vibration along each of the three mutually perpendicular axes in accordance with MIL-STD-202, Method 214, except that the test curve vibration envelope shall be as shown in Figure 1. The module shall be mounted by a method which represents the in-service mounting and the vibration shall be applied along module axes which correspond to the missile vertical (umbilical), lateral, and longitudinal axes (see 3.2.1.2.4).



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4.10.3.5 Humidity. The module shall be subjected to the humidity test in accordance with MIL-STD-202, Method 103, Test Condition B. The post-humidity tests shall be conducted after a drying period in accordance with MIL-STD-202, Method 103 (see 3.2.1.2.5).

4.10.4 Special tests.

4.10.4.1 Life expectancy test. The module life expectancy test shall be in accordance with MIL-STD-202, Method 108, Test Condition C, except the module test temperature profile and module power cycling shall be in accordance with Figure 2 (see 3.2.1.4).

4.10.4.1.1 Test chamber. The test chamber shall be in accordance with 4.5.3.

4.10.4.1.2 Module operation. The module shall be operated in accordance with the life test circuit of the applicable drawing. Module operating time can be accumulated only when power is being applied to the module.

4.10.4.1.3 Electrical characteristics. The module's electrical characteristics throughout the life test shall be in accordance with 3.2.1.1.3 and shall be periodically measured while at ambient temperature as follows:

a. Prior to starting the life expectancy test.

b. After each 20 cycles of module operating time and temperature.

c. At the conclusion of the life expectancy test (100 cycles).

The modules may be temporarily removed from the test chamber to determine its electrical characteristics.

4.10.4.2 Burn-in. Each module shall be subjected to six cycles of burn-in with thermal cycling in accordance with Figure 3. The test chamber shall be in accordance with 4.5.3. Module operation shall be in accordance with 4.10.4.1.2 (see 3.2.1.4).

4.10.5 Chamber verification test. This test shall be used to verify the chamber air temperature uniformity of 4.5.3.1. The test shall be conducted:

a. Prior to initial use of the chamber.

b. Prior to use after any repair, modification, or changes which affect air flow or thermal capacity.

c. At periodic intervals not to exceed 6 months.



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4.10.5.1 Chamber loading. The chamber verification test shall be conducted with the chamber loaded with the maximum number of modules. Simulated modules having the same dimensional and thermal characteristics may be used.

4.10.5.2 Temperature measurement points. Temperature measurements shall be taken at:

a. The eight corners (3 to 6 inches from chamber walls) and at the center of the cubic volume that encompasses all regions where test modules are placed (see Figure 4).

b. The chamber-control reference point.

4.10.5.3 Temperature sensor mounting. Each temperature sensor shall be mounted on a low thermal conductivity mount which induces a minimum disruption of air flow.

4.10.5.4 Temperature measurement. Temperature measurements shall be taken after the temperature of the test specimens has stabilized in the high temperature operating chambers and after the test specimens have stabilized at both high and low temperature in the life and burn-in test chambers. The temperature at the chamber-control reference point shall be continuously monitored for compliance with the individual test requirements. The temperature difference between the chamber-control reference point and each temperature measurement point shall be determined for compliance with 4.5.3.1. The difference shall be determined either continuously or periodically such that each measurement point is measured at least once in every 15 minutes.

4.11 Failure reporting. Unless otherwise specified in the contract or purchase order (see 6.2.1), module failures experienced during first article inspection or the final electrical test of quality conformance inspection shall be documented in accordance with AD-1322. When specified in the contract or purchase order (see 6.2.2), module production quantity, by module type, and failure report quantity shall be prepared as a monthly failure summary report.

4.12 Packaging, packing, and marking. Examination shall be made to ascertain that packaging, packing, and marking conform to Section 5 of this specification.

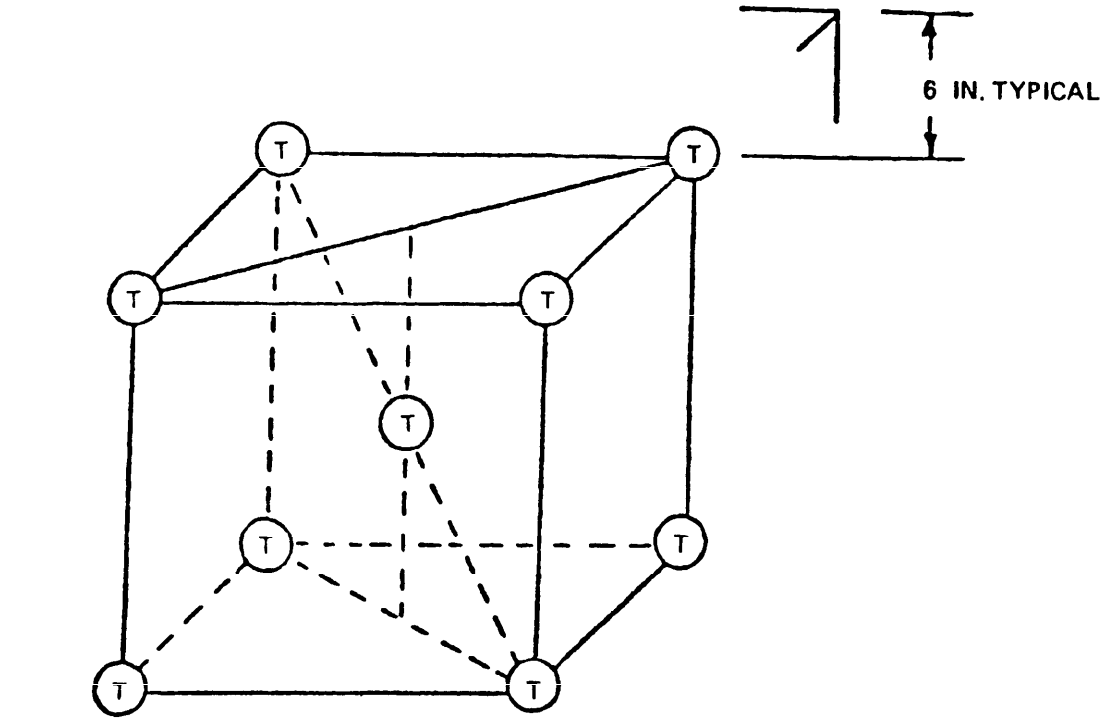
## 5. PACKAGING

5.1 Preservation and packaging. As specified in the contract or purchase order, preservation, and packaging shall be either Level A or Level C in accordance with MIL-STD-794 (see 6.2.1).

5.1.1 Level A. Modules shall be individually protected and unit packaged in accordance with MIL-P-116, Method III.



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T = TEMPERATURE MEASUREMENT DEVICE

FIGURE 4. Test chamber temperature uniformity verification measurement points.



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5.1.2 Level C. Modules shall be packaged to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Packages shall comply with the Code of Federal Regulations 49 CFR 171-178, or regulations of other carriers as applicable to the mode of transportation.

5.2 Packing.

5.2.1 Level A. Modules packaged as specified in 5.1.1, shall be packed in wood-cleated fiberboard; nailed wood; wirebound, corrugated, or solid fiberboard; wood-cleated plywood (overseas type) boxes.

5.2.2 Level C. Modules, packaged in accordance with 5.1.2, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall be in accordance with the Code of Federal Regulations 49 CFR 171-178, or regulations of other carriers as applicable to the mode of transportation.

5.3 Marking. Each module shall be marked in accordance with MIL-STD-130, with the manufacturer's name or symbol, module drawing number with revision letter, serial number, and date code, and with any other markings required on the applicable module drawings.

5.3.1 Special marking. Special marking shall be as specified in the contract or purchase order (see 6.2.1).

5.3.2 Normal marking. Unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The modules specified herein are intended for use in guided missiles.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Title and drawing number with revision letter for each module assembly ordered.

c. Responsibility for inspection and inspection facilities, if other than as specified in 4.1.



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d. Activity responsible for test station certification (see 4.5.4)

e. Minimum standards for working environments (see 4.6).

f. Test reports, if required (see 4.7.5).

g. Documentation of module failures, if different than as specified in 4.11.

h. Levels of preservation, packaging, packing, and applicable markings (see Section 5).

i. Special markings (see 5.3.1).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below should be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of Defense Acquisition Regulations (DAR) 7-104.9(n)(2) are invoked and the DD Form 1423 is not used, the data specified below should be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

| Paragraph | Data requirements | Applicable DID | Options |
|---|---|---|---|
| 4.3 | Test Procedures | UDI-T-21347 | --- |
| 4.5.1 | Certification | UDI-A-23264B | --- |
| 4.5.4 | Test Station Certification | UDI-A-23264B | --- |
| 4.7.1 | Test Plans | DI-T-5204 | --- |
| 4.7.5 | Test Report | DI-T-2072 | --- |
| 4.11 | Failure Summary Report | UDI-R-21139 | --- |

(DIDs related to this specification and identified in Section 6 should be approved and listed as such in DoD 5000.19L, Volume II, AMSDL. Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of 12 modules from each Type A module unique part number and two Type B unencapsulated modules and 12 Type B encapsulated modules from each Type B module unique part number. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, approval, and final disposition of the first article. In addition, the contracting officer should include instructions for module first article verification during production if required.



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6.3 Cognizant design activity. The cognizant design activity is the Naval Weapons Center, China Lake, California 93555.

6.4 Test reports. Test reports should include, as a minimum, the following features in addition to basic test information:

a. Purpose of test.

b. Test equipment data (e.g., types, tolerances, scales, etc.).

c. Measurements of specimens made before, during, at restart of, and at end of test, as appropriate.

d. Method of monitoring specimens electrical requirements/ characteristics during test.

e. Method of monitoring failures.

f. Abstract of test results.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.6 Supersession data. This specification supersedes the following:

a. MIL-E-85246 dated 3 June 1980.

b. MIL-E-85246, Amendment 1, dated 14 October 1982.

c. MIL-E-85246, ECP M7678R1 (NOR M7678.1A) dated 28 October 1983.

d. MIL-E-85246, ECP M8038R1 (NOR M8038.1A) dated 27 May 1983.

e. MIL-E-85246, ECP F1210R1 (NOR F1210.1A) dated 12 October 1983.

f. MIL-E-85246, ECP 11260 (NOR 11260.1) dated

Preparing activity:
Navy-AS

(Project 1420-N266)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions - Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-E-85246A(AS) | ELECTRONIC CIRCUIT MODULES |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: |

5 PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording.

c. Reason/Rationale for Recommendation.

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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